

Houdna khilouf



HuggingFace



Google Scholar



LinkedIn



ResearchGate

SUMMARY

Passionate researcher and educator specializing in Natural Language Processing (NLP), emotion analysis, and large language model (LLM) development for Arabic and Algerian dialects. Experienced in building and fine-tuning transformer-based models for emotion detection, sentiment analysis, and multimodal learning. Founder of multiple projects including DziriEmoBERT and DzBERT, pioneering efforts toward creating large-scale datasets and pre-trained models for the Algerian dialect. Skilled in Python, machine learning frameworks, and data engineering, with a strong focus on low-resource language processing and AI applications for real-world impact such as driver emotion monitoring systems. Currently pursuing a PhD in Data Science, focusing on real-time multimodal emotion recognition.

WORK EXPERIENCE

Freelancer

2022 - present

Full stack web developer & Mobile app developer

Vacation Professor

Nov 2022 - Jun 2023

National Institute of Professional Education Specializing in Professional Education (Amara Laskri), El Tarf

Specialty: Informatique, **Option:** Database

Courses: Database Fundamentals, Oracle Implementation, and Operations Research

Vacation Professor

Oct 2025 - Present

University Chadli Bendjedid El Tarf, Algeria

Department: of Computer Science **Courses:** TP/TD Algorithm and Data Structure

Department: of Law **Courses:** Cour/TP/TD Programation (Python)

EDUCATION

PhD in Data Science, University Chadli Bendjedid El Tarf, Algeria

Mar 2025 - Present

Thesis: Design and Evaluation of Multimodal Large language models for emotion analysis in low- ressource language.

Master in Intelligent Computer Systems, University Chadli Bendjedid

Oct 2022 - Jul 2024

Thesis: Multimodal Analysis of emotion for drivers.

Bachelor's degree in Computer Science, University Chadli Bendjedid

Oct 2019 - Jul 2022

Thesis: Web Application for online Library of book reviews.

LANGUAGE

Arabic: Naive, English: Reading and Speaking, French: Reading and Speaking

RESEARCH INTERESTS

Natural Language Processing, Generative Ai, Transfer Learning, Fine-Tuning, Large Language Models, Multimodal Data, Machine Learning, Deep Learning, Computer Vision, Data Science, Speech recognition, Low-resource language

Research PROJECTS

DziriEmoBERT

Fine tune the first model for the Algerian Dialect text emotions Using DziriBERT a pre-trained model for Algerian Arabic dialect and five Arabic dialects datasets. Task: text Analysis for emotion. the Model Avialabel in huggingface [Here](#)

Hackathon

May 15, 2024

Organize a hackathon on May 15, 2024, to collect a dataset for Algerian facial emotions and fine-tune this dataset using a pre-trained model for facial emotion detection.

Ameni a Startup project to reduce road accidents,

Oct 2024 -Jul 2025

In this project, we created an innovative system to reduce road accidents by developing an integrated vehicle system that uses sensors to analyze drivers' facial expressions and voice. The system automatically adjusts vehicle controls based on the detected emotional state, enhancing both safety and the driving experience. This project specifically focuses on Algerian drivers, with tailored features such as dialect recognition and adaptation to local cultural markers, including headwear such as hijabs and beards.

Dz-Emotion: Dataset for Algerian Dialect Emotion Detection

Oct2024-Jan2025

This project aims to collect a large dataset of Algerian dialect comments from social media platforms (YouTube, Facebook, Instagram,). The comments will be annotated with the correct emotions, allowing for the creation of a comprehensive dataset. This dataset will serve as the foundation for building a large language model capable of accurately recognizing and processing emotions in the Algerian dialect. (Almost completed 6 labels, each with 1000 rcomments). the Model Avialabel in huggingface [Here](#)

MEDAD Dataset

Aug, 2026

MEDAD: A Multimodal Emotion recognition for the Algerian Dialect

MEDAD is a Dataset under development for the Algerian dialect. The dataset is being collected from YouTube based in the movie With Algerian dialect. Were the videos collected and labelled for emotion recognition with six class (Happy, Sadness, disgust, fear, surprise, neutral).

DzVoice: A Low-Resource Algerian Dialect Speech Dataset

Apr 2026 - Present

DzVoice Corpus is a speech dataset developed for automatic speech recognition in the Algerian dialect, containing more 3000 aligned audio and transcription pairs (untill now). The dataset reflects real-world spoken language variability, including informal speech and code-switching. It aims to facilitate research on low-resource ASR by enabling the adaptation of pre-trained models and improving recognition performance on dialectal Arabic.

SKILLS

PROGRAMING SKILLS

Mobile Application:Flutter
Web Development: HTML, CSS ,JavaScript, PHP
Desktop Application: Delphi, Tkinter, PyQt
Database: MySQL, Firebase, Access, PostgreSQL, Oracle
Programming Languages: Python
Toolkits, Library and Packages: Transformers,Vits, librosa, NLTK, PyTorch, Keras, TensorFlow, scikit- learn, matplotlib, NumPy, Pandas, ..
Experience with: Hugging Face, Jupyter Notebooks, Kaggle, Google Colab
LaTex Vs Code , Git , GitHub... and others.

SOCIAL ACTIVITIES

Member in Itech club

2022 - present

Member in the Itech Club at Chadli Bendjedid University, focusing on artificial intelligence, web development, and mobile application development.

Member in ESP club

2023 - present

Member in the ESP Club at Chadli Bendjedid University, focusing on embedded systems and electronics.

PUBLICATIONS

- H. Khilouf, A. Ziani, N. Ahmed Malek, I. Gasmi. *DziriEmoBERT: A Pre-trained Language Model for Algerian Dialect Emotions*. Submitted to **COSI 2025** Conference accepted (indexed in springer)
- H. Khilouf, A. Ziani, D. Shwab, N. Ahmed Malek. *Dz-emotion: An Algerian Dataset for Text- Based Emotion Detection*.International Conference on Recent Advances in Mathematics and Informatics (ICRAMI), Sousse, Tunisia, 2025, pp. 1-6, doi: 10.1109/ICRAMI64946.2025.11472633.
- H. Khilouf, A. Ziani, A. Ben Khalifa, N. Ahmed Malek. Bridging the Gap for a Low-Resource Language: DzCorpus and DzBERT for Algerian Arabic Dialect, will be Submitted to an 2027 ACL Conference (RANLP;COLING;.....)
- H. Khilouf, A. Ziani, A. Ben Khalifa, N. Ahmed Malek, H. Boudjema, S. Zouiti. DzVoice: A Speech-to-Text Dataset for Algerian Dialect (in preparation and will be Submitted to MDPI *Machine Learning and Knowledge Extraction Special issue* [Advancing Natural Language Processing for Low-Resource Languages and Dialects](#))

CONFERENCE PARTICIPATION

The second edition of the International Conference on Recent Advances in Mathematics and Informatics ([ICRAMI'2025](#)), which is scheduled to take place in Sousse (Tunisia) from November 20 to 23, 2025

